Do Foreign Portfolio Investors Drive Indian Equity Returns, or Do They Simply Follow Market Momentum?

Veer Thakkar

June 2026

Abstract

Foreign Portfolio Investor (FPI) activity is frequently mentioned as a major driver of Indian equity market performance. This paper investigates the relationship between monthly FPI net flows and NIFTY 50 returns between March 2021 and March 2026. Four regression models are estimated to evaluate both contemporaneous and predictive relationships, while controlling for global risk sentiment using the CBOE Volatility Index (VIX). Results indicate a statistically significant contemporaneous association between FPI flows and NIFTY returns. However, this relationship disappears when FPI flows are lagged by one month. In contrast, the VIX emerges as a statistically significant predictor in the lagged specification. These findings suggest that foreign portfolio flows move alongside market performance but do not provide reliable predictive information regarding future returns.

1. Introduction

Foreign Portfolio Investors play a prominent role in Indian financial markets. Media commentary often attributes market rallies and declines to the buying and selling decisions of foreign investors. Consequently, a widespread belief exists that FPI flows directly influence stock market performance.

However, an important distinction exists between contemporaneous association and predictive power. While FPI flows may move together with market returns, it does not necessarily follow that foreign investor activity can predict future market movements.

This paper investigates whether FPI net flows explain NIFTY 50 returns and whether those flows possess predictive power once a one-month lag is introduced. The analysis also considers the role of global risk sentiment through the inclusion of the CBOE Volatility Index (VIX).

2. Research Question and Hypotheses

Do FPI net flows explain NIFTY 50 returns, and do they possess predictive power relative to broader measures of global risk sentiment?

H_0: FPI net flows have no statistically significant relationship with NIFTY 50 returns.

H_1: FPI net flows exhibit a statistically significant relationship with NIFTY 50 returns.

H_2: Lagged FPI flows possess statistically significant predictive power for future NIFTY 50 returns.

3. Literature Review

The relationship between foreign portfolio investment (FPI) and stock market performance has been examined in both developed and emerging markets. A central question in the literature is whether foreign investors actively influence stock prices through their trading activity or whether they primarily respond to existing market movements and economic conditions.

Research on portfolio flows into India suggests that both domestic and global factors play an important role in determining foreign investment behaviour. Gordon and Gupta (2003) find that foreign portfolio inflows are influenced by external financial conditions as well as domestic market variables, with stock market returns emerging as an important determinant of investment flows. Their findings suggest that foreign investors do not allocate capital randomly, but respond to information contained in market performance and broader economic conditions.

The direction of causality between portfolio flows and market returns remains debated. Traditional explanations often assume that foreign capital inflows affect stock prices by increasing demand for equities. However, more recent evidence challenges this one-way interpretation. Ülkü and Weber (2011) argue that the relationship between foreign trading and stock returns is more complex, showing that stock market returns can also influence subsequent foreign investment decisions. Their results suggest that foreign investors may react to market performance rather than consistently predicting it, making reverse causality an important consideration.

Another strand of literature focuses on the role of portfolio investment in market volatility. Sethi (2007) notes that portfolio flows are generally more volatile than long-term foreign direct investment and can increase fluctuations in domestic financial markets. This suggests that foreign investment may amplify short-term market movements, particularly during periods of uncertainty or heightened global risk aversion.

Overall, existing research does not support a simple conclusion that foreign portfolio investors consistently drive stock market returns. Instead, the evidence suggests a dynamic relationship in which market performance, investor behaviour, and global financial conditions interact simultaneously. The present study contributes to this discussion by examining whether monthly FPI flows possess predictive power for NIFTY 50 returns during the period March 2021 to March 2026 and by testing whether this relationship remains significant after controlling for market volatility through the VIX.

4. Data and Methodology

Monthly observations were collected for the period March 2021 to March 2026. The dependent variable is the monthly percentage change in the NIFTY 50 index. The primary explanatory variable is monthly FPI net flow measured in INR crores. The CBOE VIX is included as a proxy for global risk sentiment.

Four Ordinary Least Squares (OLS) regression models were estimated. Model 1 examined the relationship between contemporaneous FPI flows and NIFTY returns. Model 2 introduced the CBOE VIX as a control variable. Model 3 employed lagged FPI flows to test predictive power, while Model 4 incorporated both lagged FPI flows and the VIX.

Prior to estimation, correlation analysis was conducted to assess potential multicollinearity between explanatory variables. The correlation coefficient between FPI flows and the VIX was approximately -0.30, indicating only a modest inverse relationship and suggesting that severe multicollinearity is unlikely to materially affect the regression estimates.

Model Specific Methodology

Model 1:

$$Return_t = \alpha + \beta_1 FPI_t + \epsilon_t$$

Model 2:

$$Return_t = \alpha + \beta_1 FPI_t + \beta_2 VIX_t + \epsilon_t$$

Model 3:

$$Return_t = \alpha + \beta_1 FPI_{t-1} + \epsilon_t$$

Model 4:

$$Return_t = \alpha + \beta_1 FPI_{t-1} + \beta_2 VIX_t + \epsilon_t$$

Variables:
$Return_t$ = monthly NIFTY 50 return
FPI_t = monthly FPI net flow
FPI_{t-1} = one month lagged FPI net flow
VIX_t = monthly CBOE Volatility Index
α = Intercept
β = Estimated regression coefficient
ϵ_t = error term

5. Results

Table 1. Regression Results

Variable	Model 1	Model 2	Model 3	Model 4
FPI Net Flow	5.544×10^{-7}*** (0.0000015)	5.138×10^{-7}*** (0.0000014)	—	—
Lagged FPI Net Flow	—	—	-7.664×10^{-9} (0.954)	-6.08×10^{-8} (0.639)
VIX	—	−0.000992 (0.216)	—	−0.002196** (0.019)
R^2	0.326	0.344	0.00006	0.093
Adjusted R^2	0.315	0.321	−0.017	0.061
N	61	61	60	60

Notes: Values in parentheses are p-values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Model 1 examines the contemporaneous relationship between FPI net flows and NIFTY 50 returns. The coefficient on FPI flows is positive and statistically significant ($p < 0.01$), indicating that months characterised by larger foreign inflows tend to exhibit stronger equity market performance. The model explains approximately 32.6% of return variation. A ₹10,000 crore increase in FPI inflows is associated with approximately 0.55% higher monthly NIFTY returns.

Model 2 introduces the CBOE VIX as a control for global risk sentiment. The coefficient on FPI flows remains highly significant, while the VIX coefficient is statistically insignificant. This suggests that the observed relationship between FPI flows and NIFTY returns is not solely driven by global volatility conditions.

Model 3 replaces contemporaneous FPI flows with a one-month lagged variable. The coefficient becomes statistically insignificant ($p = 0.954$), and explanatory power collapses ($R^2 \approx 0.00006$). This indicates that previous-month FPI activity provides virtually no information regarding future market returns.

Model 4 incorporates both lagged FPI flows and the VIX. Lagged FPI remains insignificant ($p = 0.639$), while the VIX becomes statistically significant ($p = 0.019$). A one-point increase in the VIX is associated with approximately 0.22% lower monthly NIFTY returns. These findings suggest that global risk sentiment possesses greater predictive relevance for future NIFTY returns than historical FPI activity.

6. Discussion

The results reveal an important distinction between contemporaneous association and predictive power. While FPI net flows exhibit a strong and statistically significant relationship with NIFTY 50 returns within the same month, this relationship disappears when FPI flows are lagged by one month.

Models 1 and 2 demonstrate that periods of higher foreign capital inflows are associated with stronger equity market performance. The positive and highly significant FPI coefficients suggest that foreign investor activity and market returns move closely together. Furthermore, the persistence of FPI significance after controlling for the CBOE VIX indicates that this relationship is not solely driven by global market volatility.

However, Models 3 and 4 challenge the claim that FPI flows possess meaningful predictive power. When lagged FPI flows are used to explain future returns, the coefficients become statistically insignificant and explanatory power declines significantly. This suggests that historical FPI activity provides little information regarding subsequent market performance. The negative adjusted R^2 in Model 3 further reinforces the weakness of the lagged specification, indicating that the model performs worse than a simple prediction based on the average return.

Rather than foreign portfolio flows predicting market performance, strong market performance may attract subsequent foreign inflows. Consequently, the contemporaneous relationship observed in Models 1 and 2 should not be interpreted as evidence of a direct causal effect from FPI flows to returns.

One plausible explanation is that foreign investors respond to prevailing market conditions rather than systematically forecasting future returns. Strong market performance may attract foreign capital, creating a contemporaneous relationship without implying causality from FPI flows to future returns. Similarly, both FPI activity and equity returns may react simultaneously to common information such as macroeconomic developments, earnings expectations, monetary policy announcements, or changes in investor sentiment.

The significance of the VIX in Model 4 provides additional insight. Unlike lagged FPI flows, the VIX remained statistically significant when explaining future returns. The negative coefficient suggests that increases in global uncertainty are associated with weaker subsequent NIFTY performance. This finding is consistent with financial theory, which predicts that elevated uncertainty reduces risk appetite and leads investors to demand higher risk premiums for holding equities.

Taken together, the findings suggest that FPI flows should primarily be interpreted as a contemporaneous indicator of market conditions rather than a forecasting tool. While foreign investors remain influential participants in Indian equity markets, their trading activity appears to reflect existing market dynamics more than it predicts future movements.

7. Limitations

Several limitations should be acknowledged when interpreting the findings of this study.

First, the analysis is based on a relatively small sample of 61 monthly observations. A larger sample covering multiple market cycles may provide more robust estimates and improve statistical reliability.

Secondly, the time interval of March 2021 to March 2026 may not act as a stable economic period, due to external geo-political factors, conflicts, and economic recovery from COVID-19, directly influencing VIX and FPI flows.

Third, the use of monthly data may obscure shorter-term market dynamics. Daily or weekly observations could potentially reveal relationships that are not visible at lower frequencies.

Additionally, the study employs ordinary least squares regression and therefore identifies statistical associations rather than causal relationships. While significant coefficients indicate meaningful relationships, they do not establish that FPI flows directly cause changes in NIFTY returns.

Furthermore, the analysis includes only one control variable, namely the CBOE VIX. Other factors such as domestic institutional investor flows, interest rates, exchange rate movements, inflation expectations, corporate earnings, and monetary policy decisions may also influence market returns.

Finally, the study examines only a one-month lag structure. Alternative lag lengths may yield different results and could be explored in future research.

8. Conclusion

This paper investigated the relationship between Foreign Portfolio Investor (FPI) net flows and NIFTY 50 returns using monthly data from March 2021 to March 2026. Four regression models were estimated to evaluate both contemporaneous and predictive relationships while accounting for global risk sentiment through the CBOE Volatility Index (VIX).

The findings provide strong evidence of a contemporaneous association between FPI activity and market performance. Models 1 and 2 indicate that higher FPI inflows are associated with stronger NIFTY returns, with FPI coefficients remaining highly statistically significant even after controlling for the VIX. These results suggest that foreign investor activity is closely linked to market movements within the same period.

However, the evidence does not support the hypothesis that FPI flows possess meaningful predictive power. In Models 3 and 4, lagged FPI flows were statistically insignificant and explained virtually none of the variation in future returns. This finding indicates that historical foreign investment activity is not a reliable predictor of subsequent market performance.

In contrast, the VIX emerged as a statistically significant variable in the lagged specification. The negative relationship between the VIX and future NIFTY returns suggests that global risk sentiment contains more predictive information than past FPI flows.

Overall, the results imply that FPI flows move alongside market performance rather than systematically forecasting it. Consequently, foreign portfolio activity should be viewed primarily as a contemporaneous indicator of market conditions rather than a reliable tool for predicting future returns. Future research may extend this analysis through larger datasets, additional control variables, and alternative econometric techniques to further investigate the interaction between foreign capital flows and Indian equity markets, and investigate the reverse relationship by examining whether market returns attract subsequent foreign portfolio inflows.

References

Gordon, J., & Gupta, P. (2003). *Portfolio flows into India: Do domestic fundamentals matter?* (IMF Working Paper No. WP/03/20). International Monetary Fund.

Sethi, N. (2007). *International capital flows and growth in India: The recent experience* (SSRN Working Paper). SSRN.

Ülkü, N., & Weber, E. (2011). *Bigger fish in small pond: The interaction between foreigners' trading and emerging stock market returns under the microscope* (Working Paper No. 451). University of Regensburg.